QQ-A-200/2F
9 September 1983
SUPERSEDING
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February 20, 1981

FEDERAL SPECIFICATION SHEET

ALUMINUM ALLOY 2014, BAR, ROD, SHAPES, TUBE AND WIRE, EXTRUDED

This specification was approved by the Assistant Administrator, Office of Federal Supply and Services, General Services Administration, for the use of all Federal agencies.

The complete requirements for procuring the aluminum alloy 2014, bar, rod, shapes, tube and wire extruded described herein shall consist of this document and the latest issue of QQ-A-200/GEN (see 2.1).

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the specific requirements for aluminum alloy 2014 bar, rod, shapes, tube and wire produced by extrusion.

1.2 Classification.

1.2.1 Tempers. The bar, rod, shapes, tube and wire shall be of the following tempers: O, T4, T42, T4510, T4511, T6, T62, T6510 or T6511, as specified (see 6.2). The definition of these tempers shall be as specified in QQ-A-200/GEN.

1.2.2 Type. Tubing shall be additionally classified as follows:

Type		Description
I	-	Tubing extruded from hollow billets using die and mandrel (see QQ-A-200/GEN)
II	-	Tubing extruded from solid billets using a porthole or spider die or similar tooling (see QQ-A-200/GEN)

2. APPLICABLE DOCUMENTS

2.1 Government publications. The issues of the following documents, in effect on date of invitation for bids or solicitation for offers, form a part of this specification to the extent specified herein.

Federal Specifications

QQ-A-200/GEN - Aluminum Alloy, Bar, Rod, Shapes, Structural Shapes, Tube and Wire, Extruded; General Specification for

FSC 9530, 9525,
4710, 9540

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(Single copies of this specification, and other Federal specifications and commercial item descriptions required by activities outside the Federal Government for bidding purposes are available without charge from General Services Administration Business Service Centers in Boston, MA; New York, NY; Philadelphia, PA; Washington, DC; Atlanta, GA; Chicago, IL; Kansas City, MO; Fort Worth, TX; Houston, TX; Denver, CO; San Francisco, CA; Los Angeles, CA; and Seattle, WA.

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3. REQUIREMENTS

3.1 Chemical composition. The chemical composition shall conform to the requirements specified in table I.

TABLE I. Chemical composition 1/

Element	Percent	
	Minimum	Maximum
Copper	3.9	5.0
Silicon	0.50	1.2
Manganese	0.40	1.2
Magnesium	0.20	0.8
Iron	--	0.7
Zinc	--	0.25
Titanium	--	0.15
Chromium	--	0.10
Other elements, each	--	0.05
Other elements, total 2/	--	0.15
Aluminum	Remainder	

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TABLE I. Chemical composition 1/ - CON

1/ Analysis shall routinely be made only for the elements specifically mentioned in table I. If, however, the presence of other elements is indicated or suspected in amounts greater than the specified limits, further analysis shall be made to determine that these elements are not present in excess of specified limits

2/ The sum of those "Others" metallic elements 0.010 percent or more each, expressed to the second decimal before determining the sum

3.2 Mechanical properties.

3.2.1 Mechanical properties of material as supplied. The mechanical properties in the direction of extrusion shall conform to the requirements specified in table II.

TABLE II. Mechanical properties

Temper	Thickness, (bar and shapes); diameter, (rod and wire); wall thickness, (tube) inches	Area square inches	Tensile strength, minimum, ksi	Yield strength at 0.2 percent offset or at extension indicated		Elongation in 2 inches or 4 times D 1/, 4/, minimum, percent
				Minimum, ksi	Extension under load, inch per inch	
O	All	All	2/	2/	0.0038	12
T4, T4510, T4511	All	All	50.0	35.0	0.0053	12
T42 3/	All	All	50.0	29.0	0.0048	12
T6, T6510, T6511	Up to 0.499, incl. 0.500 to 0.749, incl. 0.750 and over 0.750 and over	All All Up thru 25 Over 25 thru 32	60.0 64.0 68.0 68.0	53.0 58.0 60.0 58.0	0.0071 0.0075 0.0077 0.0075	7 7 7 6
T62, 3/	Up to 0.749, incl. 0.750 and over 0.750 and over	All Up thru 25 Over 25 thru 32	60.0 60.0 60.0	53.0 53.0 53.0	0.0071 0.0071 0.0071	7 7 6

1/ D represents specimen diameter
2/ No minimum. Maximum tensile and yield strengths shall be 30.0 ksi and 18.0 ksi, respectively
3/ Material in the T42 or T62 tempers is not available from the material producers
4/ See QQ-A-200/GEN for elongation requirement exceptions

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3.2.2 Mechanical properties after heat treatment. In addition to conforming to the requirements of 3.2.1, materials identified in the following paragraphs shall, after having been heat-treated to other tempers also identified therein, have properties in the extrusion direction conforming to those specified in table II, as applicable.

3.2.2.1 Material in the O temper. Material in the O temper, without the subsequent imposition of cold work or forming operations, shall, after proper solution heat treatment and natural aging, develop the properties specified in table II for the T42 temper (see 6.2 and 6.3).

3.2.2.2 Material in the T4, T4510, T4511, T6, T6510 and T6511 tempers. Material in the T4, T4510, T4511, T6, T6510 and T6511 tempers, without the subsequent imposition of cold work or forming operations, shall be capable of being re-solution treated and naturally aged to the properties specified for the T42 temper. Such capability shall be demonstrated when specified (see 6.2 and 6.3).

3.2.2.3 Material in the T4, T4510 and T4511 tempers. Material in the T4, T4510 and T4511 tempers shall be age-hardenable artificially to the properties specified for the T6, T6510 and T6511 tempers, respectively. Such capability shall be demonstrated when specified (see 6.2).

3.2.2.4 Material in the T42 temper. Material in the T42 temper shall be age-hardenable artificially to the properties specified for the T62 temper. Such capability shall be demonstrated when specified (see 6.2).

3.3 Internal defects. When specified (see 6.2), rod, bar and shapes shall be ultrasonically inspected (see QQ-A-200/GEN). Acceptance limits shall be as specified in table III.

TABLE III. Ultrasonic discontinuity acceptance limits 1/ 3/

Thickness, (bar and shapes) (inches) 4/	Maximum weight of piece (pounds)	Maximum width to thickness ratio	Discontinuity class 2/
0.500 and over	600	10 to 1	B

1/ Discontinuities in excess of those listed in table III may be allowed subject to the approval of the procuring activity, if it is established that they will be removed by machining or that they are in noncritical areas

2/ See QQ-A-200/GEN

3/ Requirements for ultrasonic inspection for sizes, maximum weights and discontinuity classes differing from those shown in table III and for rod shall be specified in the contract provided the techniques and standards are agreed to by the procuring activity and the producer (see 6.2)

4/ For thickness of a shape, see definition for shape in QQ-A-200/GEN

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3.4 Marking. In addition to marking required by QQ-A-200/GEN, metal in the T6, T6510 and T6511 tempers and, when specified (see 6.2), metal in the T4, T4510 and T4511 tempers, shall also be identified by lot number, marked in at least one location on each piece.

4. QUALITY ASSURANCE PROVISIONS (see QQ-A-200/GEN)

4.1 Heat treatment.

4.1.1 Aging treatment before testing. Specimens selected for treatment to T4, T42, T4510 and T4511 tempers may be tested within 4 days after completion of the solution heat treatment. If these tests fail, the manufacturer may elect to test additional specimens from the same sample lot after the expiration of at least 4 days after the date of solution heat treatment. These specimens shall be taken from the same location in the lot as the prior specimens.

4.2 Mechanical test after heat treatment.

4.2.1 Number of tests after heat treatment. From material in each temper of those specified for heat treatment capability demonstrations in 3.2.2 and 6.2, an additional number of specimens equal to that required in QQ-A-200/GEN shall be taken and tested after heat treatment to each temper specified to determine conformance to 3.2.2.

5. PREPARATION FOR DELIVERY (see QQ-A-200/GEN)

6. NOTES

6.1 Intended use. This alloy is intended for use where high strength is required, usually no welding is involved, and where optimum corrosion resistance is not required. The T4510 and T6510 tempers are intended primarily to provide extruded material with low residual stresses and consequent minimum distortion during machining. The T4511 and T6511 tempers are straightened T4510 and T6510 material, respectively, and may have unknown residual stresses and may or may not distort during machining.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number and date of this specification
(b) Form, quantity and temper required (see 1.2.1)
(c) Dimensions required
(d) Mechanical properties and dimensional tolerance requirements for sizes not specifically covered
(e) Demonstrations of heat treatability of material in tempers other than the O temper (see 3.2.2 and 4.2.1)

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(f) Whether ultrasonic inspection of bar, rod and shapes is required (see 3.3)
(g) Special end use requirements
(h) Selection of applicable levels of preservation, and packing, whether level A, level B or commercial (see QQ-A-200/GEN)
(i) Whether lot numbering for material in the T4, T4510 or T4511 tempers is required
(j) Whether type I tubing is required (see 1.2.2)

6.3 Properties after heat treatment. Mechanical properties and corrosion properties, where applicable, are certified for the temper of material supplied. The producers capability demonstration is not evidence that user-treated material conforms to property requirements of a given temper. Frequently, user-heat-treated material may develop a lower level of properties, especially if any cold, warm or hot work is introduced, prior to solution heat treatment. The user should be held responsible for demonstrating that his processing will yield properties meeting requirements.

MILITARY INTEREST:

Custodians

Army-MR
Navy-AS
Air Force-20

Review Activities

Army-AR, EA, MI
DLA-IS

User Activities

Army-CR
Navy-MC, SH, YD

CIVIL AGENCY COORDINATING ACTIVITIES:

GSA-FSS
NASA-JFK, MSF
DOE-BPA

PREPARING ACTIVITY:

NAVY-AS

DOD Project 9530-0226

Orders for this publication are to be placed with the General Services Administration, acting as an agent for the Superintendent of Documents. See Section 2 of this specification to obtain extra copies and other documents referenced herein.

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